

BipedNav: monocular stair perception and virtual-holonomic-constraint walking for a knee-less biped re-implementation, verification and stress test of the 2024 perception poster and the ECE1658 controller

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Abstract

BipedNav couples two pieces of earlier work: a perception layer that recovers staircase geometry (rise h , run b) from a single RGB image through a monocular metric-depth network and plane extraction (Qiu, Narayanan, Cheng, Laschowski, 2024), and a locomotion layer that walks a planar two-link “acrobot” biped with one hip actuator by enforcing a virtual holonomic constraint (VHC) $q_2 = \phi(q_1)$ through feedback linearisation (ECE1658 course project, 2024, originally in MATLAB). Both layers were re-implemented in Python from the archived code and data, tested, and extended: the perception pipeline now identifies treads and risers from surface normals and clusters plane offsets, reaching a mean absolute rise error of 1.7 cm (RealSense cloud) and 2.1 cm (monocular cloud) against tape measurements on the five archived scenes; the walker is simulated as a hybrid system on real terrain with collision checking of the swing foot against treads and risers. The flat-ground design of the report is reproduced to 4×10^{-5} in the polynomial coefficients and its measured return-map slope (0.9329) matches the analytic Floquet multiplier $\delta^2/M^- = 0.9326$. Three findings are new. (i) The impact map of the course template relabels velocities by a reflection $R = -I$, which is not the Jacobian of the leg swap; with the kinematically exact relabelling the same gait retains $\delta = 0.115$ instead of 0.336 of its phase velocity and is no longer a stable limit cycle, although the optimiser still finds exact-map gaits with swing-foot retraction. (ii) A point-foot, equal-leg, knee-less biped can only pass its legs with a vertical stance leg without dipping into a flat tread, which requires an aperture $\beta > 2\alpha$ for stair pitch α and confines the landing point to within $b - \sqrt{2h - h^2}$ (leg units) of the nosing; human-scale stairs ($\alpha \approx 23^\circ$) are out of reach for this morphology, and the report’s stair proof of concept relied on the slope model that hides a 5 cm dip at the rear nosing. (iii) Within the reachable regime, the same VHC gaits designed for a slope walk real collision-checked staircases when the foot placement obeys a simple rule: land within 5% of the tread depth from the nosing when ascending (5° pitch), stand on the nosing when descending (5° and 10°).

Keywords: monocular depth estimation, stair parameter extraction, virtual holonomic constraints, hybrid zero dynamics, acrobot biped, impact map, stair climbing.

1 Overview

The archived project (BipedNav/) contains the perception workspace (RealSense ROS driver, bag processing scripts, five sample scenes with RGB, RealSense depth, monocular depth/normal estimates and hand measurements) and the control code (MATLAB `control/MATLAB/PART1`, a partial Python port). Nothing in the archive could be executed as is: the MATLAB scripts need the Symbolic Math Toolbox session state, the Python port was unfinished, and the perception scripts assume a ROS Noetic install. This document records the re-implementation in `experiments/`:

- `stair_perception.py`: staircase geometry from organised point clouds (Section 2.1);
- `acrobot_vhc.py`: model, impact map, VHC design (linear and optimised), constrained dynamics, terrain model, hybrid simulation (Section 2.2);
- `design_all.py`, `sweep_stairs.py`: gait design batches; `run_experiments.py`, `fig_perception.py`: every figure and number in this document; `make_video.py`: the demonstration video; `render.py`: plotting in the website’s visual language.

The web page <https://thejerrycheng.github.io/bipednav.html> presents the same material.

2 Method

2.1 Perception: stair geometry from one image

The 2024 poster fine-tuned Metric3D v2 [1] on 16 780 RGB-D stair images, back-projected the metric depth with the camera intrinsics and extracted planes by iterative RANSAC. The archived sample set contains, per scene, the RealSense D435 depth (640×480 px, mm), the organised point clouds from that depth (“RealSense cloud”, binary PLY) and from the network’s depth (“monocular cloud”, ASCII PLY), the network’s normal map ($3 \times 480 \times 640$), and two tape measurements of rise and run per staircase (`measurements.csv`). The estimated-depth tensors are empty files in the archive; the monocular clouds are used instead.

Iterative RANSAC on these clouds fragments the grazing treads and fits planes across tread/riser edges (the first re-run gave 6–9 cm “risers” on three scenes), so the pipeline was rebuilt around the poster’s own idea of predominant normals:

1. *Normals*. Central differences on the image grid (baseline 3 px, jumps > 6 cm rejected), oriented toward the camera; every second pixel is kept ($\approx 7 \times 10^4$ points).
2. *Predominant normals*. A 4° spherical histogram of the normals; its modes are refined as means over 18° cones. The tread normal $\mathbf{u}$ is the most populated mode pointing up (negative camera- y). The walking direction $\mathbf{f}$ is the optical axis projected on the tread plane, refined by the riser mode if a mode exists $\geq 70^\circ$ from $\mathbf{u}$ and within 30° of $-\mathbf{f}$ (this rejects side walls, whose normals are lateral).
3. *Refinement*. $\mathbf{u}$ is re-estimated as the least-squares normal of the most populated tread cluster with a slab shrinking from 4 cm to 1.2 cm.
4. *Offset clustering*. Tread points ($\angle(\mathbf{n}, \mathbf{u}) < 15^\circ$) are projected on $\mathbf{u}$, riser points ($\angle(\mathbf{n}, -\mathbf{f}) < 20^\circ$) on $\mathbf{f}$; a 5 mm histogram smoothed by a Gaussian ($\sigma = 1$ cm) gives the plane offsets as peaks at least 7 cm apart and above 15% of the maximum.
5. *Lattice fit*. Rise h is the least-squares lattice spacing of the tread offsets (adjacent gaps between 6 and 45 cm); run b is the lattice spacing of the riser offsets (15–60 cm) or, when fewer than two risers are visible, of the front edges (5th percentile along $\mathbf{f}$) of consecutive treads.

Camera pitch follows from $\mathbf{u}$: $\tan \psi = -u_z/u_y$. The whole pipeline is deterministic and runs in 0.09 s (RealSense) / 0.32 s (ASCII read dominated) per scene on a laptop CPU.

2.2 Acrobot model and impact map

Two identical legs of length $l = 1$ m, mass $m = 1$ kg at $l_c = l/2$, inertia $I_z = ml^2/12$, one torque τ at the hip. q_1 is the absolute angle of the stance leg (foot to hip) from the $+x$ axis and q_2 the relative angle of the swing leg; $B = [0 \ 1]^\top$, $B^\perp = [1 \ 0]$. The pinned dynamics $D(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = B\tau$

are closed-form,

$$D = \begin{bmatrix} ml_c^2 + m(l^2 + l_c^2 + 2ll_c \cos q_2) + 2I_z & m(l_c^2 + ll_c \cos q_2) + I_z \\ \cdot & ml_c^2 + I_z \end{bmatrix}, \quad (1)$$

$$C = \begin{bmatrix} \eta \dot{q}_2 & \eta(\dot{q}_1 + \dot{q}_2) \\ -\eta \dot{q}_1 & 0 \end{bmatrix}, \quad \eta = -mll_c \sin q_2, \quad (2)$$

$G = mg[(l + l_c) \cos q_1 + l_c \cos(q_1 + q_2), l_c \cos(q_1 + q_2)]^\top$, and were verified against the sympy derivation of the MATLAB template to 9×10^{-16} (`Model.check_closed_forms`). The impact is the rigid, plastic impact of Westervelt et al. [3]: with the unpinned mass matrix $\bar{D}(q)$ of coordinates (q, x) and $E = [\partial p_{\text{sw}} / \partial q \ I_2]$,

$$\begin{bmatrix} \bar{D} & -E^\top \\ E & 0 \end{bmatrix} \begin{bmatrix} \dot{q}^+ \\ F \end{bmatrix} = \begin{bmatrix} \bar{D} [I_2 \ 0]^\top \dot{q}^- \\ 0 \end{bmatrix}, \quad \dot{q}_{\text{post}} = \Delta_{\dot{q}}(q^-) \dot{q}^-, \quad (3)$$

which was checked to 10^{-15} against conservation of angular momentum about the new stance foot and of the old stance leg about the hip. The legs are then relabelled. The exact swap is $q_1^+ = q_1 + q_2 - \pi$, $q_2^+ = 2\pi - q_2$, whose Jacobian is $R_{\text{kin}} = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$. The course template instead uses $T(q) = -q + [\pi + 2\alpha, 2\pi]^\top$ and $\dot{q}^+ = -\dot{q}_{\text{post}}$, i.e. $R_{\text{refl}} = -I$. On the impact surface $q_2 + 2q_1 = 2\pi + 2\alpha$ both give the same configuration, but the two velocity maps agree only for velocities tangent to that surface ($\dot{q}_2 = -2\dot{q}_1$), which post-impact velocities are not (the former stance foot lifts, so $J_{\text{sw}} \dot{q}^+ = -\dot{x}^+ \neq 0$). Physically the new stance leg keeps the angular velocity the old swing leg had, $\dot{q}_1^+ = \dot{q}_1 + \dot{q}_2$; the reflection assigns it $-\dot{q}_1$ instead. Both conventions are implemented (`Model(relabel=...)`) and every result states which one it uses; the impact matrix of the handout is $\mathcal{I} = R \Delta_{\dot{q}}(q^-)$.

2.3 VHC design

Following the ECE1658 handout, $q = \sigma(\theta) = [q_1^+ - \theta \tilde{q}_1, \phi(\theta)]^\top$ with $\theta \in [0, 1]$, $\tilde{q}_1 = \beta$, $q^\pm = [(\pi \pm \beta)/2 + \alpha, \pi \mp \beta]^\top$ for aperture β and (design) slope α , and ϕ a polynomial of degree $k - 1$ satisfying

$$\phi(0) = q_2^+, \quad \phi(1) = q_2^-, \quad \phi(\theta_f) = \pi, \quad \phi'(0) = f(v_1), \quad \phi'(1) = v_1, \quad \phi'(\theta_f) = v_2, \quad (4)$$

where $f(v_1) = -\tilde{q}_1(-\mathcal{I}_{21}\tilde{q}_1 + \mathcal{I}_{22}v_1)/(-\mathcal{I}_{11}\tilde{q}_1 + \mathcal{I}_{12}v_1)$ makes $\sigma'(0) \parallel \mathcal{I}\sigma'(1)$ (hybrid invariance). The report uses $\theta_f = 1/2$; Section 2.6 explains why treads need $\theta_f = 1/2 + \alpha/\beta$. For $k = 6$ the six conditions determine ϕ (a 6×6 Vandermonde solve); for $k > 6$ and for slopes and stairs the design is the nonlinear program of report Part 2/4, solved here with SLSQP after eliminating the equalities analytically: the decision vector is $(v_1, v_2, c_6, \dots, c_{k-1})$ and the six lowest coefficients follow from (4). Inequalities: transversality margin $\min_\theta (B^\perp D\sigma')^2 \geq \zeta$ without sign change; existence $\delta^2/M^- \leq 0.95$; $V^- < 0$ and the energy margin $-V^- \delta^2/(M^- - \delta^2) \geq 1.05 V_{\text{max}}$; for flat ground the no-scuff tangent bounds $v_1 < 2\tilde{q}_1$, $f(v_1) < 2\tilde{q}_1$, $v_2 > 2\tilde{q}_1$; and, sampled at 241 phases, positive clearance of the swing foot above the terrain (treads, risers or slope line) except at the three legitimate contacts. The objective is the curve length $\int \|\sigma'\| d\theta$ with a 10^{-5} ridge on the coefficients. A Nelder–Mead penalty polish runs when SLSQP stops infeasible.

2.4 Constrained dynamics, stability test and controller

With $\Psi_1 = -B^\perp G/(B^\perp D\sigma')$ and $\Psi_2 = -B^\perp (D\sigma'' + C\sigma')/(B^\perp D\sigma')$ evaluated on σ , the zero dynamics is $\ddot{\theta} = \Psi_1 + \Psi_2 \dot{\theta}^2$, with virtual mass $M(\theta) = \exp(-2 \int_0^\theta \Psi_2)$ and potential $V(\theta) =$

$-\int_0^\theta \Psi_1 M$ (cumulative trapezoid on 801 points). With $\delta = \langle \sigma'(0), \mathcal{I}\sigma'(1) \rangle / \|\sigma'(0)\|^2$ the restricted Poincaré map in $\zeta = \frac{1}{2}M^-(\dot{\theta}^-)^2$ is affine, $\zeta_{k+1} = (\delta^2/M^-)\zeta_k - V^-$, so the limit cycle exists and is exponentially stable iff $0 < \delta^2/M^- < 1$, $V^- < 0$ and $-V^-\delta^2/(M^- - \delta^2) > V_{\max}$, with $\dot{\theta}^{+*} = \delta\sqrt{-2V^-/(M^- - \delta^2)}$ and Floquet multiplier δ^2/M^- . The controller is the report’s input–output linearisation of $y = q_2 - \phi(\theta(q_1))$,

$$\tau = (HD^{-1}B)^{-1} \left(HD^{-1}(C\dot{q} + G) - K_p \sin y - K_d H\dot{q} - \dot{H}\dot{q} \right), \quad H = [\phi'/\tilde{q}_1, 1], \quad K_p = 40^2, \quad K_d = 80. \quad (5)$$

2.5 Terrain and collision model

A staircase is the set of treads $n \in \mathbb{Z}$ at height nh spanning $x \in [d + (n-1)b, d + nb)$ relative to the stance foot, so d is the distance from the stance foot to the next riser and $h < 0$ descends; $h = 0$ is flat ground. Its design slope is $\alpha = \arctan(h/b)$ and its aperture $\beta = 2 \arcsin(\sqrt{b^2 + h^2}/2l)$; a slope is the line $y = x \tan \alpha$ with stride s . One step is integrated in two phases (DOP853 via `solve_ivp`, `rtol` 10^{-9}): until the swing foot passes $x = d$ (checking on the dense output that it never goes below the floor except within 5% of l of the stance foot, where the legs legitimately fold), then until it touches the landing surface; a foot below the next tread when it crosses the riser plane is a *trip*, a floor violation deeper than 1 mm a *scuff*, a hip lower than $0.25l$ a *fall*. Then (3) and the relabelling are applied at the actual touchdown configuration.

2.6 Why stairs are hard for a knee-less point-foot biped

Two facts of pure geometry. (a) The swing leg passes the stance leg only in the folded configuration $q_2 = \pi$, where the swing foot coincides with the stance foot and moves perpendicular to the legs. On a flat tread the foot therefore dips into the step unless the stance leg is vertical at that instant, $q_1 = \pi/2$, which happens at $\theta_f = \frac{1}{2} + \alpha/\beta$ and requires $\beta > 2|\alpha|$; the slope model of the report ($\theta_f = \frac{1}{2}$, $q_1 = \pi/2 + \alpha$) lets the foot cross the stance point along the slope line, i.e. $(b-d)\tan\alpha$ inside the tread behind the stance foot (Fig. 10). (b) The swing foot moves on a circle of radius l around the hip, whose height is at most l (vertical stance leg), so the foot can only be above the next riser, at horizontal distance d , if $l - \sqrt{l^2 - d^2} \geq h$, i.e. $d \geq \sqrt{2h - h^2}$ in leg units, while landing on that tread needs $d \leq b$. Fig. 9 maps the admissible landing window $(b - \sqrt{2h - h^2})/b$ over stair pitch and aperture. Human stairs seen by the perception layer (rise 12 cm, run 28 cm, $\alpha = 23^\circ$) need $\beta > 46^\circ$, i.e. a stride of $0.79l$ (2.6 steps at once for a 1 m leg, or a 0.38 m leg for one step at a time), and even then a window of a few centimetres. The report’s proof-of-concept staircase ($b = 1$, $h = 0.5$, $d = 0.9$) sits exactly at the boundary: window $0.134b$, $\theta_f = 0.89$.

3 Experiments

All numbers are produced by `python3 experiments/run_experiments.py` and `fig.perception.py` and stored in `experiments/out/metrics.json`, `perception_results.json`, `gaits.json`. Seeds are fixed; there is no randomness in the controller pipeline.

3.1 Perception

Table 1 and Fig. 1 summarise the result. Rise is recovered to 1.7 cm MAE from the RealSense cloud and 2.1 cm from the monocular cloud; run to 2.5 cm from both. The largest errors are the WE 17:51:34 scene (camera pitched only 47° , risers seen at grazing angle: the riser lattice locks

Table 1: Stair geometry from the five archived scenes versus the tape measure (mean of the two measured steps). MAE over the five scenes; the poster (iterative RANSAC, its own five test cases) reported 2.91 cm / 5.1 cm.

scene	tape h cm	D435 h cm	mono h cm	tape b cm	D435 b cm	mono b cm	pitch deg
WE 17:51:34	14.2	16.5	10.7	27.1	19.3	25.9	47
WE 18:27:47	19.5	19.6	19.9	25.3	25.1	23.3	67
WI 17:41:57	12.0	15.1	15.5	29.6	31.2	33.5	56
WI 17:47:26	12.0	14.2	12.0	29.6	28.1	28.3	53
WI 18:12:19	15.0	15.7	18.0	32.1	33.3	27.7	53
MAE		1.7	2.1		2.5	2.5	
accuracy		86.7 %	84.9 %		91.2 %	91.4 %	

onto a 19 cm harmonic on the D435 cloud while the tread-edge estimate gives 25.6 cm) and the WI 17:41:57 scene, where both clouds return a 3 cm larger rise than the tape (the treads there are worn concrete with a raised nosing strip). The monocular cloud is metrically consistent with the RealSense cloud on all scenes (pitch within 8° , offsets within 3 cm), which is the point of the poster.

3.2 Flat ground: reproduction and verification of the report

With $\beta = 0.316637$, $v_1 = -0.894373$, $v_2 = 1.9$ and the reflection impact map, the linear design returns $a = [2.8250, 0.4246, -7.3543, 30.1847, -35.9442, 13.3225]$ (ascending powers), identical to the report to 4.4×10^{-5} , and $\mathcal{I} = \begin{bmatrix} -0.6864 & 0.3621 \\ -0.1459 & -0.1081 \end{bmatrix}$, $\delta = 0.3364$, $M^- = 0.1214$, $V^- = -0.2543$, $V_{\max} = 3.420$, $\min_\theta |B^\perp D\sigma'| = 0.100$, hybrid-invariance residual 2×10^{-14} , $\dot{\theta}^{+*} = 2.652$, $\dot{\theta}^{-*} = 7.883$ (Fig. 3–Fig. 4). Simulated from the limit cycle the walker takes 25 steps with period 1.044 s, stride 0.3153 m, speed 0.302 m/s, hip torque 110 Nm peak / 13.5 Nm rms, $|y| < 3 \times 10^{-11}$, 8.9 J of actuator work per step, and $\dot{\theta}^-$ constant to 2×10^{-3} . Started at $1.2 \times$ and $1.6 \times \dot{\theta}^{+*}$ the measured return-map slope is 0.9329 and 0.9327 against the analytic $\delta^2/M^- = 0.9326$ (Fig. 5); this slope is the Floquet multiplier, so a velocity error decays by only 7% per step and 25 steps remove 83% of it, consistent with the slow convergence visible in the report’s phase portraits. The energy margin at the fixed point is only 0.09 above $V_{\max}$: started at $0.8 \times \dot{\theta}^{+*}$ the walker stalls before the apex, and the on-manifold sweep (Fig. 6) shows the basin is one-sided, $\dot{\theta}_0/\dot{\theta}^{+*} \in [1.0, 3.0]$ (25 steps completed, upper end of the sweep). Off the manifold (both joints offset by ϵ at $1.2 \times \dot{\theta}^{+*}$) the walker survives $\epsilon \in [-0.26, 0]$ rad and fails for any positive offset. The report’s statements on disturbances (Section F) only tested faster initial conditions, which is the tolerant side.

3.3 Impact-map convention

Table 2 compares the two relabellings for the report’s parameters. Under the exact leg swap the new stance leg keeps $-0.036 \dot{\theta}^-$ instead of the reflection’s $-0.107 \dot{\theta}^-$ (δ drops from 0.336 to 0.115), and the gait fails the energy test by a wide margin: no $k = 6$ design is feasible for $\beta \in \{0.2, 0.32, 0.45, 0.6, \pi/4\}$ over $v_1 \in [-1.6, 2\beta]$, $v_2 \in [2\beta, 6]$ (45×45 grid). Retention improves with swing-leg retraction ($v_1 > 0$, Fig. 7) and the optimiser finds exact-map flat gaits with $k = 8$ at $v_1 \rightarrow 2\tilde{q}_1$ (the foot touches down with zero vertical velocity), $\delta^2/M^- \approx 0.90$; they are marginal and were not used further. On the -10° slope, however, the exact map yields a comfortable retraction

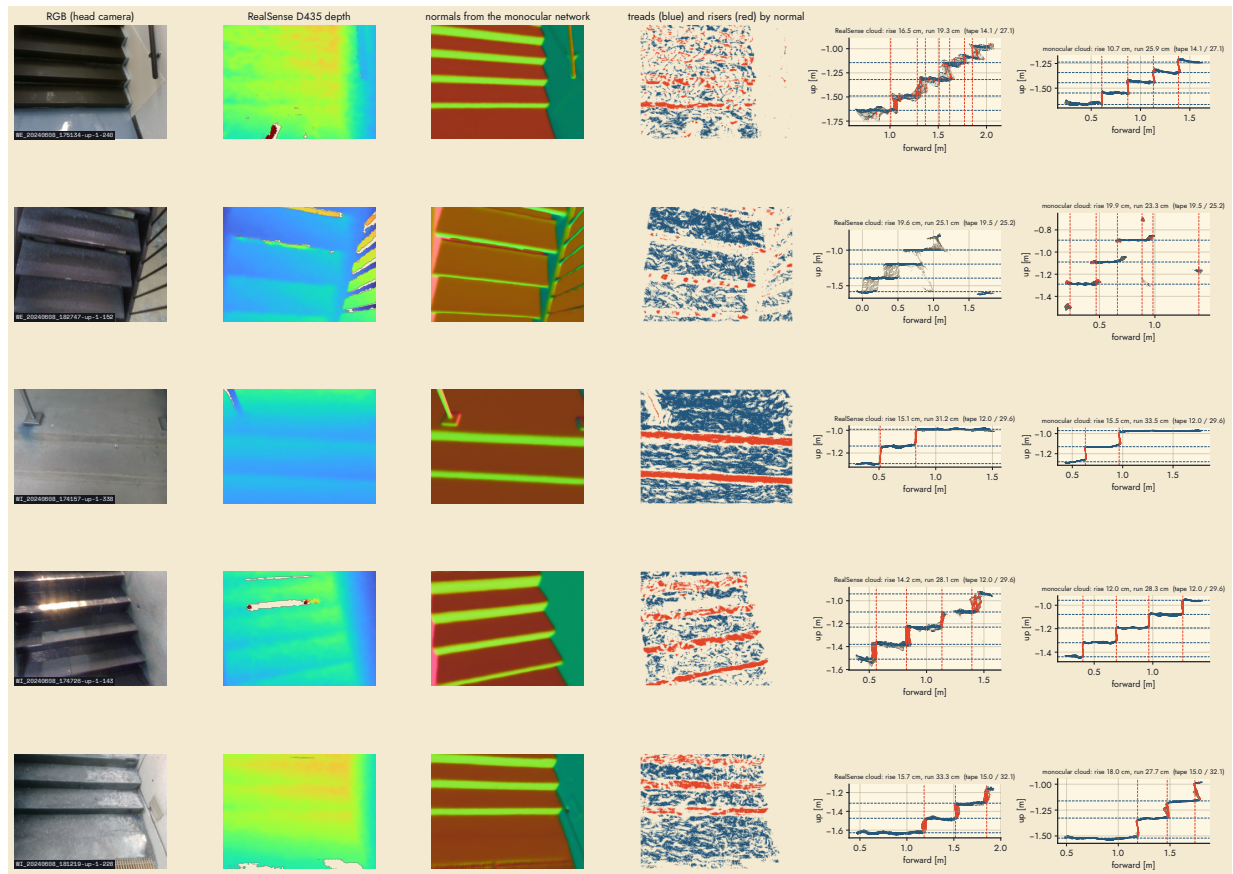


Figure 1: Per scene: RGB, RealSense depth, monocular normals, tread/riser labels from the normals, and the side profile of a 0.5 m band of the cloud with the fitted tread (blue) and riser (red) lattices, for the RealSense and the monocular cloud.

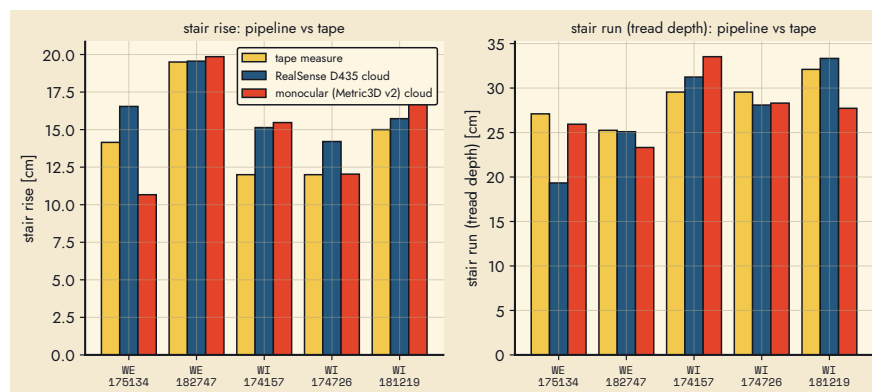


Figure 2: Rise and run per scene: tape, RealSense cloud, monocular cloud.

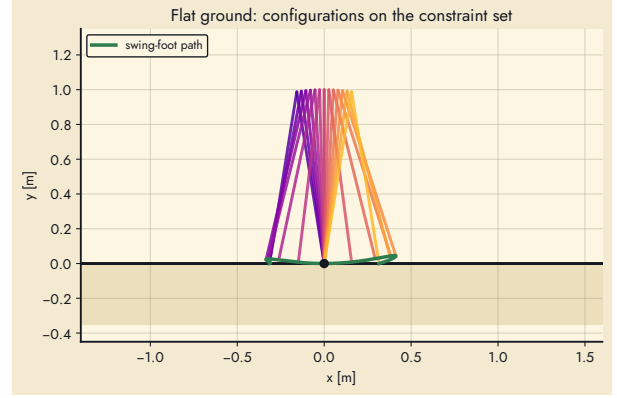
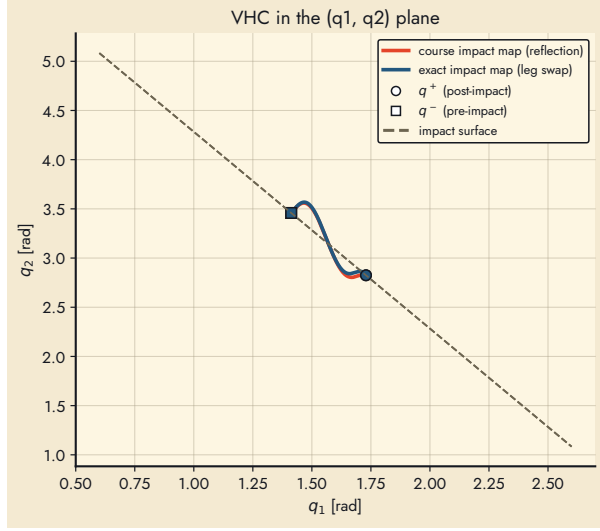


Figure 3: Left: the VHC in the (q_1, q_2) plane for the report's parameters under the reflection impact map (red) and the exact leg swap (blue); the exact map moves $\phi'(0)$ from 0.42 to 1.25. Right: the constraint set as a strobe with the swing-foot path.

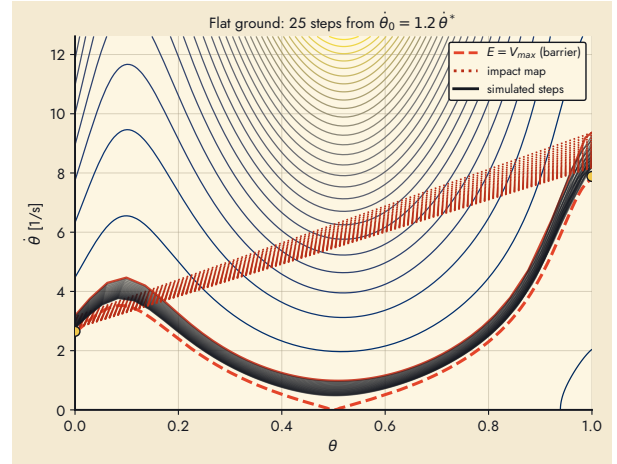
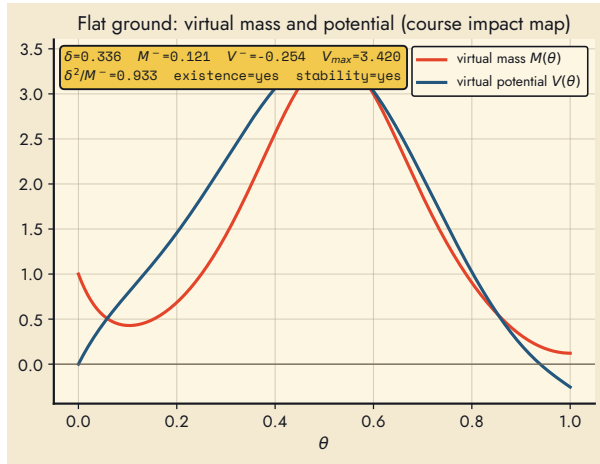


Figure 4: Left: virtual mass and potential of the report design. Right: energy levels of the constrained dynamics with 25 simulated steps from $1.2\dot{\theta}^{+*}$ (first step red, impact map dotted, fixed point yellow).

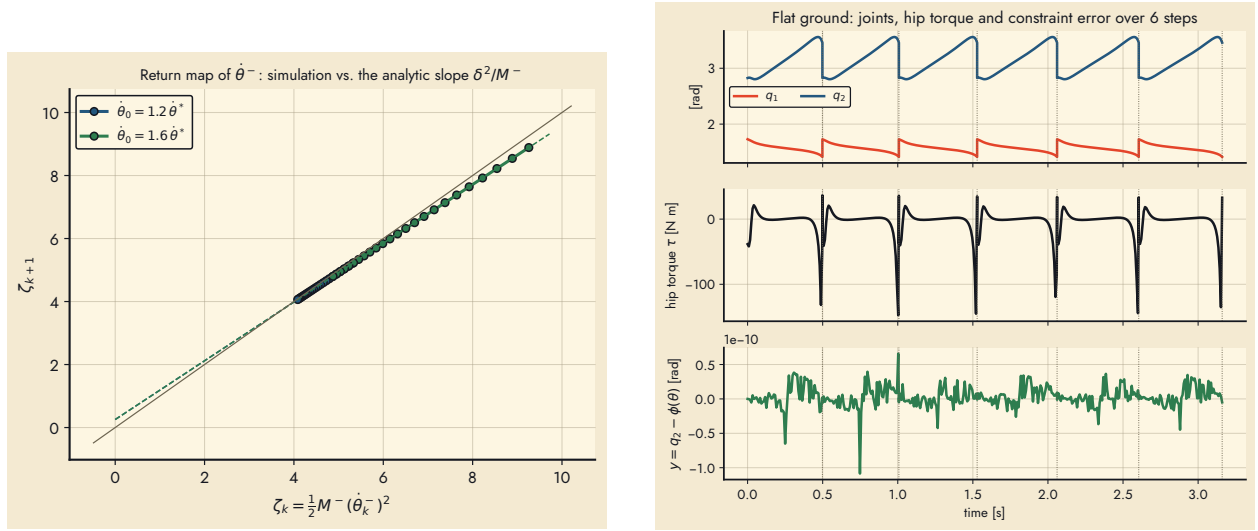


Figure 5: Left: measured return map of ζ_k for two initial speeds against the analytic affine map. Right: joint angles, hip torque and constraint error over six steps.

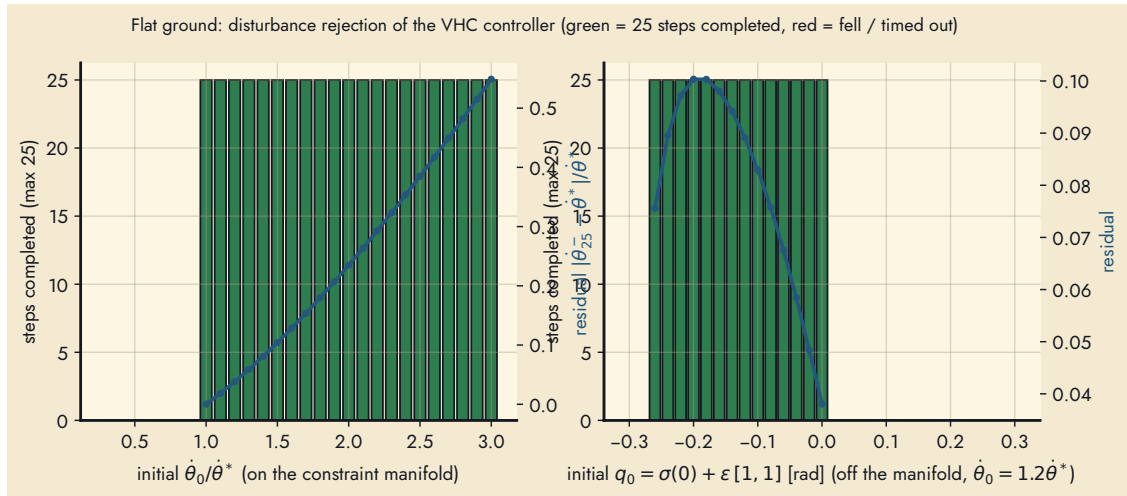


Figure 6: Disturbance sweeps on flat ground (25-step budget). Left: initial phase velocity on the manifold. Right: joint offset off the manifold.

gait ($v_1 = 1.36$, multiplier 0.44, hip torque 12 N m peak against 204 N m for the reflection gait, Table 3).

Table 2: Report parameters ($\beta = 0.3166$, $v_1 = -0.8944$, $v_2 = 1.9$) under the two velocity relabellings.

relabelling	$\mathcal{I}_{11}$	$\mathcal{I}_{12}$	δ	M^-	V^-	$V_{\max}$	δ^2/M^-	stable orbit
reflection $R = -I$ (course)	-0.686	0.362	0.336	0.121	-0.254	3.42	0.933	yes
exact leg swap R_{kin}	0.832	-0.254	0.115	0.061	-0.203	1.67	0.214	no

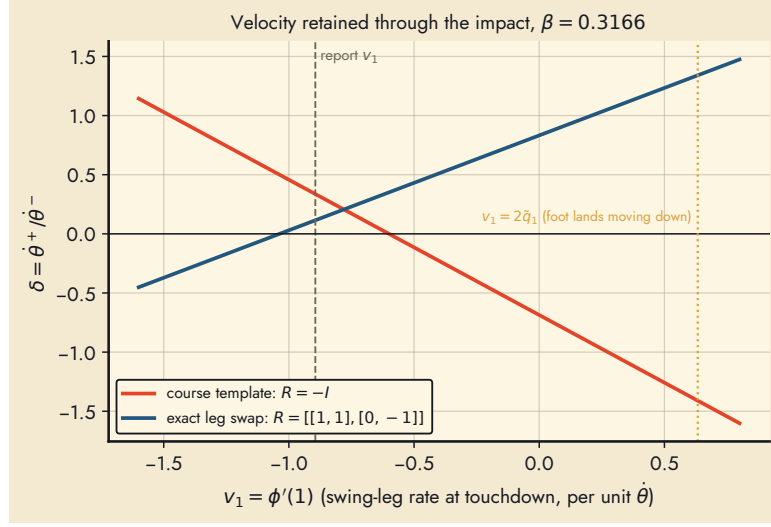


Figure 7: Phase-velocity retention δ through the impact as a function of the swing-leg rate at touchdown, for both conventions.

3.4 Slopes

Gaits were optimised for $\beta = \pi/4$ as in report Part 3 with $\alpha \in \{-10, -5, +5, +10\}^\circ$ from seven starts, $k \in \{6, 8\}$, and accepted only if they walk 25 steps in the collision-checked simulation (Table 3, Fig. 8). Downhill gaits are found for both conventions; uphill only $+5^\circ$ under the reflection ($k = 8$, multiplier 0.95, i.e. barely stable), and none at $+10^\circ$: the virtual potential V^- that the impact has to balance becomes positive as the slope rises, and the polynomial family cannot make it negative while clearing the ground. The report's $+10^\circ$ and $+30^\circ$ cases could not be reproduced with the stability test enforced; the report shows their VHC curves and strobes but no return map.

3.5 Stairs

Geometry. Fig. 9 is the map of Section 2.6; Fig. 10 shows a slope-model curve for the report's staircase against the real treads: the swing foot is 4.8 cm inside the step at $\theta = 0.46$ (prediction $(b - d) \tan \alpha = 5.0$ cm).

Optimised stair gaits. With the vertical-fold condition $\theta_f = \frac{1}{2} + \alpha/\beta$ and full tread/riser clearance, 80 cells $\{5, 10, 15, 20, 26.6\}^\circ \times \{45, 60, 68, 80\}^\circ \times \{\text{up, down}\} \times \{\text{reflection, exact}\}$ with

Table 3: Slope gaits ($\beta = \pi/4$, stride 0.765 m), 12 simulated steps from $1.15 \dot{\theta}^{+*}$.

case	map	k	v_1	v_2	δ^2/M^-	period [s]	speed [m/s]	τ_{peak} [N m]	steps
-10°	refl.	6	-2.02	1.72	0.380	1.03	0.75	204	12/12
-10°	exact	6	+1.36	1.76	0.443	1.00	0.78	12	12/12
-5°	refl.	6	-2.55	1.74	0.575	0.93	0.85	242	12/12
$+5^\circ$	refl.	8	-3.32	3.29	0.950	0.59	1.29	90	12/12
$+10^\circ$	both	—	no feasible design (14 starts per convention)						

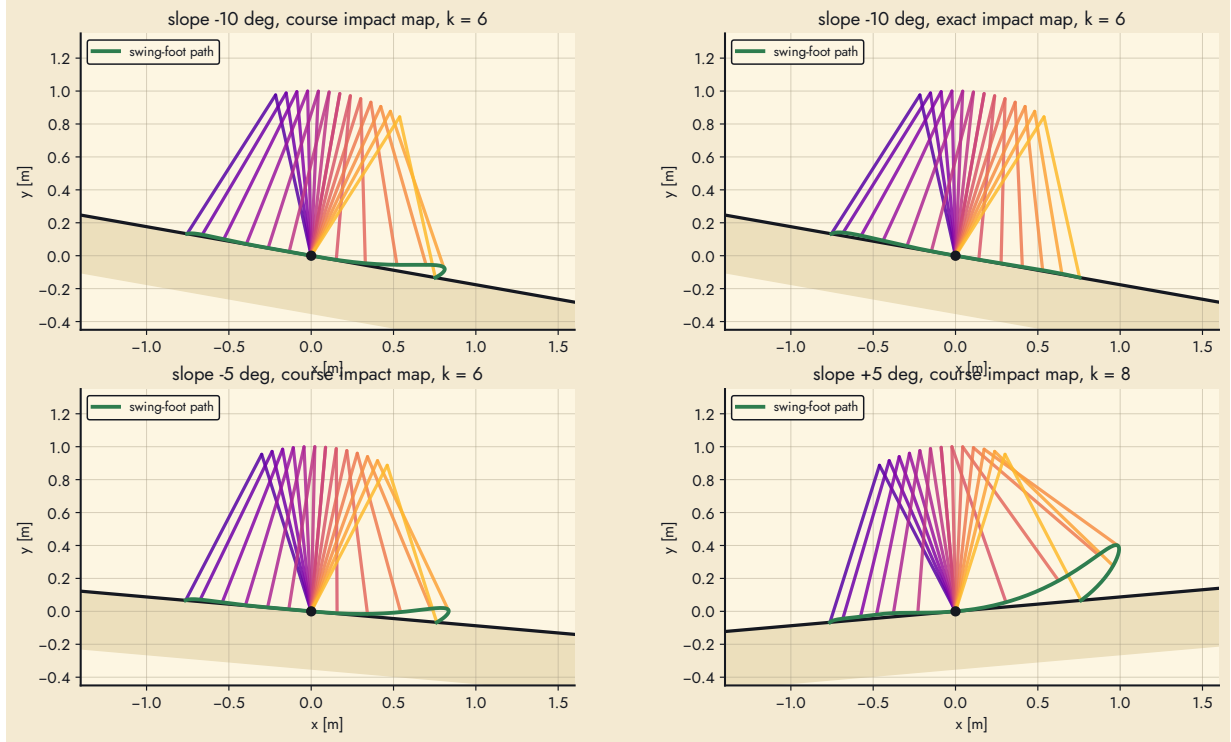


Figure 8: Constraint sets of the four slope gaits with their swing-foot paths.

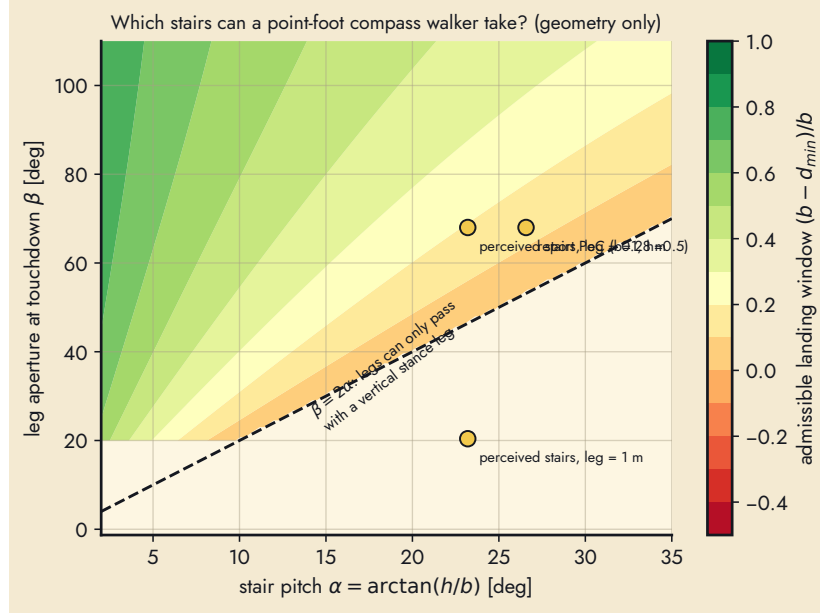


Figure 9: Admissible landing window $(b - d_{\min})/b$ for a point-foot compass walker versus stair pitch and touchdown aperture; white where the legs cannot pass with a vertical stance leg ($\beta \leq 2\alpha$).

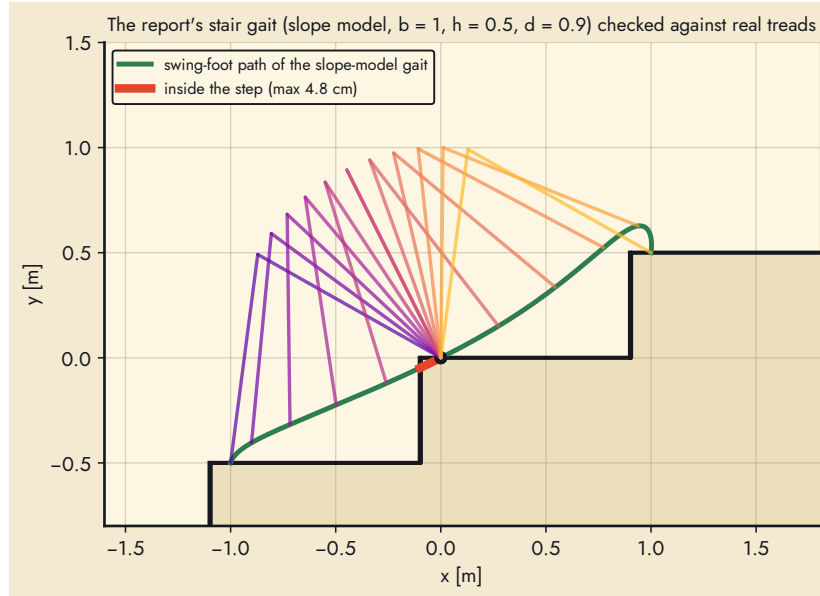


Figure 10: The slope model's swing-foot path on the report's staircase ($b = 1$, $h = 0.5$, $d = 0.9$): the highlighted part of the path is inside the step behind the stance foot.

$d = 0.8b$ and 10 starts each (`sweep_stairs.py`) produced no feasible design, nor did the batch for the report’s staircase and for the perceived staircase scaled to a 0.28 m leg (`design_all.py`). The binding constraints are the vertical-fold condition together with the riser clearance, which force the swing leg to travel almost the whole stride while the stance leg stays within a few degrees of vertical; the resulting ϕ' breaks transversality or the energy test.

Slope gaits on real staircases. The slope gaits of Table 3 were then run on staircases of the same pitch and stride (rise 6.7 cm, run 76 cm at 5° ; rise 13.3 cm, run 75 cm at 10°) while sweeping the foot placement d/b (Fig. 11, Table 4). Ascending works iff $d/b \geq 0.95$: landing within 5 % of the tread depth from the nosing puts the rear nosing so close behind the stance foot that the fold’s dip stays within the 1 mm scuff tolerance, and the leading foot clears the next riser because it arrives from above with retraction. Descending works iff $d/b \leq 0.05$: standing on the nosing, the swing foot leaving the fold downward is already past the tread. Both rules are direct consequences of Section 2.6. In the accepted placements the walker takes 12/12 steps with zero collisions; ascending at 1.02 m/s (climb rate 0.089 m/s, 77 N m peak), descending at 0.71 m/s (189 N m reflection, 9.3 N m exact map).

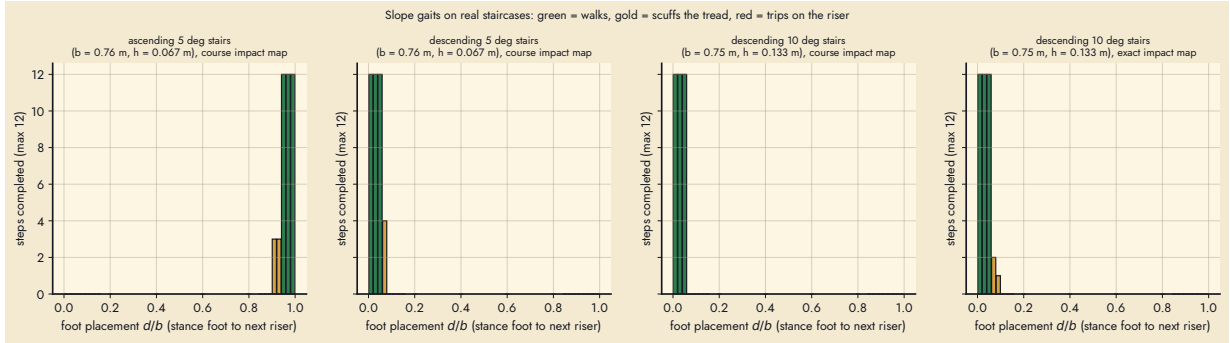


Figure 11: Slope gaits on real staircases: steps completed (of 12) versus foot placement d/b .

Table 4: Slope gaits on collision-checked staircases at the chosen foot placement.

stairs	b / h [m]	gait (map)	d/b ok	used	period [s]	speed [m/s]	τ_{peak} [N m]	steps
up 5°	0.76 / 0.067	+ 5° (refl., $k=8$)	[0.95, 0.99]	0.97	0.75	1.02	77	12/12
down 5°	0.76 / 0.067	− 5° (refl.)	[0.01, 0.05]	0.03	0.99	0.77	217	12/12
down 10°	0.75 / 0.133	− 10° (refl.)	[0.01, 0.05]	0.03	1.07	0.71	189	12/12
down 10°	0.75 / 0.133	− 10° (exact)	[0.01, 0.05]	0.03	1.04	0.73	9.3	12/12

4 Implementation and Usage

```
cd BipedNav/experiments
python3 stair_perception.py out/perception_results.json # 10 clouds, ~2 s
python3 fig_perception.py                               # figures + perception_summary.json
python3 design_all.py                                   # gait batch -> out/gaits.json (2 min)
python3 sweep_stairs.py                                 # 80-cell stair feasibility sweep
python3 run_experiments.py                              # flat impact geometry slopes stairs
python3 make_video.py                                   # out/video/bipednav_demo.mp4
```

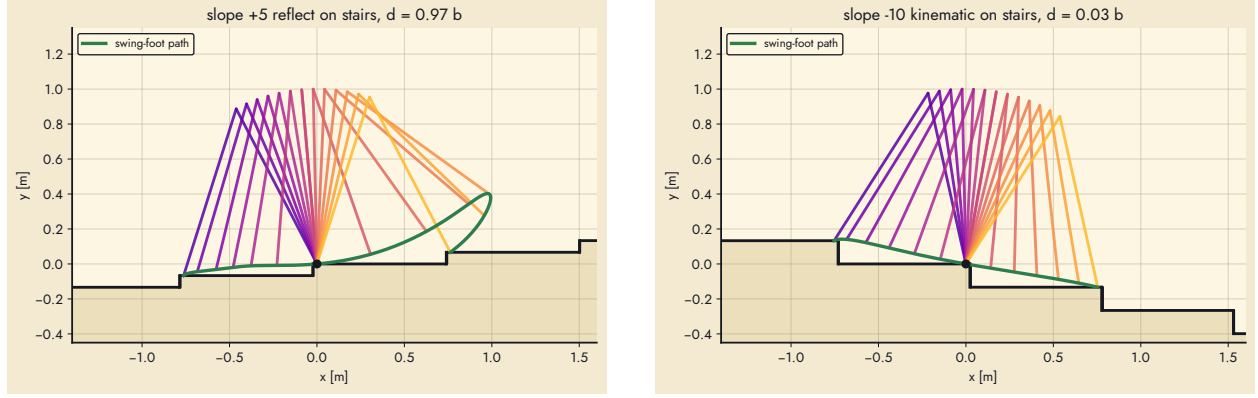


Figure 12: Left: ascending the 5° staircase with $d = 0.97b$. Right: descending the 10° staircase with the exact-map gait and $d = 0.03b$.

```
cd ../docs && make
```

```
# this document
```

`acrobot_vhc.py` composes as `Model` (dynamics, impact) $\rightarrow$ `Gait` (design + analysis) $\rightarrow$ `Terrain` $\rightarrow$ `simulate`. Physical parameters live in `Params`; a real robot would replace `controller()`'s model terms with measured ones and `simulate()` with the hardware loop, keeping `Gait` unchanged. The MATLAB originals remain in `control/MATLAB` for reference; MATLAB itself was not required for any result (the batch launcher of R2023b returned no output on this machine). Fonts for the figures are the site's (Limelight, Bangers, Jost, Space Mono) under `experiments/fonts`.

5 Limitations

- Five scenes are too few for a statistically meaningful perception evaluation; the tape measurements were taken within minutes of the recordings but not at the exact steps in view, and the poster's own test set is not in the archive. The pipeline assumes a camera pitched down $40\text{--}70^\circ$ and visible risers or tread edges.
- The walker is a planar, point-foot, knee-less, torso-less model with massless actuation and a plastic impact; torques of 100 N m – 240 N m for a 2 kg mechanism are not realistic, and the flat-ground gait's Floquet multiplier of 0.93 and one-sided basin make it fragile.
- The impact-convention question (Section 2.2) changes the quantitative conclusions of the report; the exact swap is derived from the kinematics and verified numerically, but the course material could define the acrobot walker differently, and this should be confirmed with the instructor before publication.
- Stair climbing at human pitch is infeasible for this morphology (Section 2.6); the demonstration uses 5° and 10° staircases with 75 cm treads and a foot-placement rule that a real controller would have to enforce through step-to-step foot targeting, which is outside the VHC framework as implemented.
- The gait optimiser is local (SLSQP from a handful of starts) and the stair sweep is coarse; infeasibility claims are “no design found under these constraints”, not proofs.

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